SOFTWARE REQUIREMENTS SPECIFICATION

Store Management System

VERSION 1.0

20 October 2026

Functional and non-functional requirements for a secure, reliable, multi-store management system.

REFERENCE SCENARIO

Amiable Retail Ltd

Stores in Lagos, Abuja, and Kaduna

REQUIREMENTS

30 functional

FR-01 – FR-30

30 non-functional

NFR-01 – NFR-30

STATUS

Proposed

Service levels to confirm before production commitments

PREFACE

A coherent multi-store system

You have a strong starting point. The main improvement I would make is to combine your original 30 functional requirements and 30 non-functional requirements with the six implementation sections, so the requirements describe one coherent, multi-store Store Management System rather than a generic e-commerce website.

Your system is primarily designed for business owners managing multiple stores, store managers, cashiers, inventory officers, and finance staff. Therefore, the requirements should prioritize accurate stock control, sales, purchasing, inter-store transfers, employee permissions, financial records, and reporting.

I'll keep the requirements measurable and testable. Each requirement will have a unique ID, a clear title, and bullet points stating what the system must do or what test it must pass.

A key distinction: functional requirements describe what the system does; non-functional requirements describe how well, securely, or reliably it must do it. I'll also distinguish mandatory requirements from optional capabilities where necessary, so your developers don't accidentally build a much larger e-commerce platform than you need.

INTRODUCTION

How to read this specification

Document	Software Requirements Specification
System	Store Management System
Version	1.0 · Proposed · 20 October 2026
Audience	Business owners, store managers, cashiers, inventory officers, and finance staff

Document purpose

Define the functional and non-functional requirements for a secure, reliable, multi-store management system.

Requirement convention

“Shall” indicates a mandatory requirement. Each requirement has a unique identifier and measurable acceptance criteria.

Shall	A mandatory requirement. The system is not complete if a “shall” is unmet.
Acceptance criteria	The observable result that shows a functional requirement has been met.
Verification	The test or evidence used to check a non-functional target.
Conditional	A “where”, “if”, or “optional” statement applies only when that condition is named. It is not a reason to build a larger platform than this specification asks for.

Initial reference scenario

> Amiable Retail Ltd operates stores in Lagos, Abuja, and Kaduna. The system shall support additional businesses, stores, users, products, and transactions without requiring a redesign of its core data model.

In scope

Day-to-day operation of the business and its stores.

- Business and store setup
- User access control
- Product management
- Inventory
- Point-of-sale sales
- Customers
- Suppliers
- Purchasing
- Transfers
- Expenses
- Returns
- Reporting
- Notifications
- Audit logs

Out of scope

For the initial release, unless separately approved.

- Online shopping storefront
- Customer social login
- Loyalty programs
- Courier integrations
- Product recommendations
- Advanced promotional campaigns

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Sixty requirements. Functional items are FR-01 to FR-30. Non-functional items are NFR-01 to NFR-30.

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PART A

Functional requirements

FR-01 – FR-30

Functional requirements describe what the system does. “Shall” indicates a mandatory requirement. Each requirement has a unique identifier and measurable acceptance criteria.

PART A · SECTION 1

1. Business and Store Management

FR-01

Business Registration and Profile Management

- The system shall allow an authorized owner or administrator to create and manage a business profile containing its legal name, trading name, registration number where applicable, business address, contact details, and logo.
- Each business shall have a unique system-generated identifier.
- The system shall allow authorized users to edit business information and record the user and timestamp associated with each change.
- The system shall prevent users from accessing or modifying another business’s records unless explicitly authorized.

> ACCEPTANCE CRITERIA
> An authorized owner can create, view, and update a business profile; unauthorized users cannot change it or access another business’s data.

FR-02

Store Creation and Management

- The system shall allow authorized users to create, view, edit, activate, and deactivate stores under a business.
- Each store shall have a unique identifier, store name, address, contact details, status, and business association.
- The system shall display a list of all stores belonging to the authorized business and support searching by store name or identifier.
- A deactivated store shall not accept new sales, stock movements, purchases, or transfers, except explicitly authorized historical corrections or administrative operations.
- Existing historical records shall remain accessible according to user permissions.

> ACCEPTANCE CRITERIA
> The owner can create three stores, edit their details, and deactivate one without deleting its historical transactions.

FR-03

Business and Store Settings

- The system shall allow authorized users to configure the business's base currency, applicable tax settings, receipt information, timezone, and business contact details.
- The system shall allow store-level settings such as receipt numbering, transaction prefixes, and other approved operational preferences.
- The system shall validate settings before saving and prevent invalid currency codes, tax rates, and required fields.
- Changes affecting financial calculations shall apply according to a documented effective-date policy and shall not silently rewrite historical transactions.

> ACCEPTANCE CRITERIA
> A manager with permission can update permitted store settings; a cashier cannot change tax settings or the business's base currency.

PART A · SECTION 2

2. Accounts and Access Control

FR-04

Authentication and Session Management

- The system shall allow authorized users to log in and log out using their registered credentials.
- It shall provide password change and forgotten-password recovery through a verified recovery method.
- Passwords shall never be stored or logged in plaintext.
- The system shall reject invalid, deactivated, and unauthorized accounts from protected endpoints.
- Sessions or tokens shall support expiration, revocation, and logout according to the configured security policy.

> ACCEPTANCE CRITERIA
> Unauthenticated requests to protected endpoints return an appropriate unauthorized response, and a revoked session cannot continue performing protected actions.

FR-05

Employee and Role Management

- The system shall allow authorized administrators to create, edit, deactivate, and reactivate employee accounts.
- Staff accounts shall be created by authorized personnel rather than through unrestricted public registration.
- The system shall support, at minimum, the roles of Business Owner, Administrator, Store Manager, Cashier, Inventory Officer, and Finance Officer.
- Authorized administrators shall be able to assign employees to one or more stores and configure permissions according to their assigned roles.
- The same employee shall be allowed to hold different roles at different stores.

> ACCEPTANCE CRITERIA
> An employee can be a manager in Lagos and a cashier in Abuja, with the permissions for each action determined by the active store context.

FR-06

Business and Store-Level Authorization

- Every protected request shall validate the authenticated user, business association, store assignment, role, and required permission.
- The server shall derive authorization from trusted identity and access records rather than trusting a store ID supplied by the client.
- Users shall only access records belonging to their authorized business and assigned stores.
- Unauthorized cross-store or cross-business requests shall be rejected and logged.

> ACCEPTANCE CRITERIA
> A Lagos cashier who changes a request's store ID to Abuja shall receive a forbidden response and shall not read or modify Abuja records.

PART A · SECTION 3

3. Product Management

FR-07

Product Catalogue Management

- Authorized staff shall be able to create, view, edit, archive, and search products.
- Each product shall support a unique SKU, product name, category, description where applicable, cost price, selling price, unit of measurement, and active or archived status.
- Barcode, brand, image, and additional attributes shall be supported where applicable.
- The system shall validate required fields and prevent duplicate SKUs within the defined business scope.
- Products referenced by historical transactions shall be archived rather than permanently deleted.

> ACCEPTANCE CRITERIA
> A product can be created, updated, searched, and archived; its historical sales and stock movements remain readable.

FR-08

Categories, Variants, and Product Identification

- The system shall allow authorized staff to create, edit, and archive product categories.
- Products shall support optional variants such as size, colour, model, or packaging, where each variant requires independently managed stock.
- Each independently stocked variant shall have a unique SKU and may have a unique barcode, cost price, and selling price.
- The system shall support barcode-based product lookup and manual SKU or product-name searches.

> ACCEPTANCE CRITERIA
> Two variants of the same product can be identified separately, sold separately, and tracked using their respective stock records.

FR-09

Pricing and Price History

- The system shall maintain selling prices and cost prices for products or variants.
- Authorized staff shall be able to change prices subject to role permissions and any configured approval thresholds.
- The system shall record previous and new prices, the effective date, the user making the change, and the timestamp.
- A completed sale shall retain the actual unit price and discount applied at the time of sale, regardless of subsequent price changes.

> ACCEPTANCE CRITERIA
> Changing a product's selling price shall affect eligible future sales but shall not alter the prices or totals of completed historical sales.

PART A · SECTION 4

4. Inventory and Stock Control

FR-10

Store-Level Inventory Tracking

- The system shall maintain an independent stock balance for every stocked product or variant at each store.
- Stock shall be measured using the product's defined unit of measurement.
- Authorized staff shall be able to view current stock, reserved stock if reservations are enabled, and available stock for each store.
- The system shall support inventory searches and filters by product, SKU, category, store, and stock status.

> ACCEPTANCE CRITERIA
> A product can have 20 units in Lagos, 15 in Abuja, and 8 in Kaduna, with each balance displayed independently.

FR-11

Centralized Stock Movement Processing

- Every stock increase or decrease shall pass through one shared, validated inventory transaction service.
- The system shall record stock movements resulting from sales, purchases, returns, transfers, adjustments, and write-offs.
- Each stock movement shall include the product, store, quantity, movement type, reason or source transaction, timestamp, and responsible user or system process.
- Stock updates and their corresponding movement records shall be committed atomically.

> ACCEPTANCE CRITERIA
> Every successful stock change produces a matching movement record; a failed transaction leaves neither an incorrect balance nor a partial movement record.

FR-12

Stock Adjustments and Physical Counts

- Authorized staff shall be able to perform stock adjustments with a mandatory reason and supporting reference where applicable.
- The system shall support physical stock counts and compare counted quantities against recorded quantities.
- Differences shall be recorded as proposed adjustments and shall require approval when the configured threshold or policy demands it.
- Every approved adjustment shall update stock and create an auditable movement record.
- The system shall prevent negative stock unless an explicitly approved business policy permits it.

> ACCEPTANCE CRITERIA
> A count discrepancy is visible before approval, and the stock balance changes only after an authorized adjustment is committed.

FR-13

Low-Stock Monitoring

- The system shall allow a reorder level to be configured for each product or variant at each store.
- The system shall identify products whose available stock is at or below their configured reorder level.
- Authorized users shall be able to view low-stock items filtered by store, category, and product.
- The system shall create a low-stock notification when a stock change causes a product to cross the configured threshold.
- Duplicate notifications for the same unresolved low-stock condition shall be controlled.

> ACCEPTANCE CRITERIA
> When stock falls to or below its reorder level, the product appears on the low-stock list and an eligible notification is generated.

FR-14

Inventory Batches, Expiry, and Damaged Goods

- Where required by the product type, the system shall record batch or lot numbers, expiry dates, and quantities per batch.
- The system shall identify expired and near-expiry stock using a configurable warning period.
- Where batch tracking applies, stock issued for sale shall follow first-expiry-first-out allocation unless an authorized policy specifies otherwise.
- Damaged, expired, or unsellable goods shall be recorded separately from saleable stock.

> ACCEPTANCE CRITERIA
> The system can identify a product batch approaching expiry and prevent expired stock from being sold when the applicable sales policy prohibits it.

PART A · SECTION 5

5. Sales, Returns, and Customers

FR-15

Point-of-Sale Transactions

- Authorized cashiers shall be able to search for products, scan barcodes, add items to a sale, change quantities, remove items, and review the transaction total.
- The system shall calculate line totals, applicable discounts, taxes, and the final amount payable using defined rounding rules.
- Each sale shall have a unique transaction identifier and record its store, cashier, timestamp, items, quantities, unit prices, discounts, taxes, and total.
- Completed sales shall reduce stock through the centralized inventory transaction service.

> ACCEPTANCE CRITERIA
> Selling two bags of rice from a Lagos balance of 100 reduces that balance to 98 and creates one complete sale record.

FR-16

Payment Recording and Reconciliation

- The system shall support recording cash, card, bank transfer, and other configured payment methods.
- A sale shall support one payment method or split payments across multiple methods.
- Each payment record shall include its amount, method, status, transaction reference where applicable, and association with the sale.
- The system shall distinguish pending, successful, failed, reversed, and refunded payments where applicable.
- Sales shall not be marked fully paid until the successful recorded payments satisfy the amount due, subject to explicitly configured credit or pay-later policies.

> ACCEPTANCE CRITERIA
> A split payment is accepted only when its valid payment allocations satisfy the applicable sale balance, and payment totals can be reconciled against sales.

FR-17

Receipts, Invoices, and Sales History

- The system shall generate a receipt for every completed sale using a unique receipt or transaction number.
- Receipts shall include the business and store details, transaction date, cashier, items, quantities, prices, discounts, taxes where applicable, payment methods, and amount paid.
- The system shall support printing and reprinting receipts subject to permissions and shall identify reprints where required.
- Authorized users shall be able to search and view sales by date, store, cashier, transaction ID, and payment status.
- Historical receipts shall reproduce the original transaction values.

> ACCEPTANCE CRITERIA
> A completed sale generates a retrievable receipt whose totals match the saved sale and payment records.

FR-18

Sales Returns and Refunds

- Authorized staff shall be able to locate an original sale and initiate a full or partial return against its eligible items and quantities.
- The system shall enforce the configured return period, return eligibility rules, and quantity limits.
- Returned items shall be classified according to their condition and whether they may re-enter saleable stock.
- Refunds shall be linked to the original sale and recorded separately with their amount, method, status, reason, and approval details where required.
- The system shall prevent duplicate refunds or returns exceeding the eligible quantity or refundable amount.

> ACCEPTANCE CRITERIA
> Returning one of two purchased units cannot return more than one unit, and any stock increase occurs only when the item is eligible for restocking.

FR-19

Customer Records

- Authorized staff shall be able to create, view, edit, and search customer records.
- Customer records shall support a name, optional contact details, and other approved business information.
- A customer may be associated with multiple sales, and authorized users shall be able to view the customer's purchase history.
- The system shall support anonymous or walk-in sales without requiring a full customer profile.
- Access to customer information shall be restricted according to permissions and applicable privacy rules.

> ACCEPTANCE CRITERIA
> An authorized user can retrieve a customer's eligible purchase history, while a cashier can complete a walk-in sale without creating a customer account.

PART A · SECTION 6

6. Purchasing, Suppliers, and Transfers

FR-20

Supplier Management

- Authorized staff shall be able to create, edit, search, view, and deactivate supplier records.
- Supplier records shall support a unique identifier, name, contact information, address where applicable, products supplied, and payment terms where supported.
- The system shall link each purchase order, goods receipt, and purchase return to the relevant supplier.
- Authorized users shall be able to view supplier transaction history and outstanding balances when credit purchasing is enabled.

> ACCEPTANCE CRITERIA
> A supplier's purchase history can be retrieved without combining transactions belonging to another supplier.

FR-21

Purchase Orders and Goods Receiving

- Authorized staff shall be able to create, edit before approval, submit, approve, and cancel purchase orders according to permissions.
- A purchase order shall contain a supplier, destination store, products, quantities, unit costs, and applicable additional charges.
- The system shall support full and partial goods receipts and track quantities ordered, received, outstanding, and cancelled.
- Only confirmed goods receipts shall increase store inventory.
- The system shall prevent receiving quantities beyond the outstanding purchase quantity unless an authorized exception is recorded.

> ACCEPTANCE CRITERIA
> Receiving 100 units of rice into Lagos increases Lagos stock by 100 and records the receipt against the corresponding purchase order.

FR-22

Inter-Store Stock Transfers

- Authorized staff shall be able to request, approve, reject, dispatch, receive, and cancel eligible stock transfers.
- Each transfer shall record its source store, destination store, products, quantities, status, requesting user, approving user where required, and timestamps.
- A transfer shall not be completed if the source lacks sufficient available stock.
- The system shall maintain a documented in-transit state where dispatch and destination receipt occur at different times.
- Stock shall not be counted simultaneously as available at both stores during a transfer.

> ACCEPTANCE CRITERIA
> A transfer of 10 phones from Lagos to Abuja reduces available Lagos stock and increases Abuja stock only at the appropriate recorded stages; a rejected transfer does not change stock.

FR-23

Purchase Returns and Supplier Balances

- Authorized staff shall be able to initiate and record returns of received goods to suppliers.
- Each purchase return shall reference the original receipt or purchase transaction and include items, quantities, reasons, and financial values.
- Confirmed returned goods shall reduce the appropriate inventory balance through the stock movement service.
- Where supplier credit or outstanding balances are supported, the system shall update the relevant payable or supplier credit records.
- The system shall prevent returning more goods than were received and remain eligible for return.

> ACCEPTANCE CRITERIA
> A purchase return of five units reduces stock by five only after confirmation and creates a linked supplier return record.

PART A · SECTION 7

7. Expenses, Reporting, Notifications, and Audit

FR-24

Store Expense Management

- Authorized staff shall be able to create, edit before approval, submit, approve, reject, and view store expense records.
- Each expense shall include an amount, currency, category, store, date, description, payment status, and supporting reference or attachment where required.
- Expense categories shall be configurable by authorized users.
- Expenses shall be filterable by date, store, category, status, and recording user.
- Changes and approvals shall be recorded in the audit log.

> ACCEPTANCE CRITERIA
> An approved electricity expense appears in the correct store's expense report and is excluded from approved-expense totals while still pending or rejected.

FR-25

Operational Dashboard and Reports

- The system shall provide authorized users with dashboards and reports for sales, payments, inventory, stock movements, purchases, transfers, expenses, returns, and product performance.
- Reports shall be filterable by date range and store; applicable reports shall also support product, category, supplier, cashier, and transaction filters.
- Business owners shall be able to view aggregated results across authorized stores and drill down into individual stores.
- Store managers shall only see information for stores they are authorized to access.
- Reports shall distinguish sales revenue, collected payments, refunds, expenses, inventory value, and profit-related measures rather than treating them as interchangeable.

> ACCEPTANCE CRITERIA
> Report totals reconcile with the underlying records for the same filters and accounting definitions.

FR-26

Notifications and Operational Alerts

- The system shall generate in-app notifications for low-stock conditions, transfer requests, transfer approvals or rejections, purchase approvals, and other configured events.
- Notifications shall identify the event, relevant store, creation time, and appropriate action or record reference.
- Notifications shall be delivered only to authorized recipients based on role, store assignment, and event type.
- The system shall track notification status, including unread, read, and failed delivery where applicable.
- Optional email or SMS delivery shall use a configurable provider and shall not cause a completed sale to be rolled back solely because notification delivery failed.

> ACCEPTANCE CRITERIA
> An authorized destination-store manager receives a notification for a relevant transfer event and cannot access unrelated stores through that notification.

FR-27

Activity Logs and Audit Trail

- The system shall record security-sensitive and business-critical actions, including logins where required, permission changes, product and price changes, stock movements, sales, refunds, purchases, transfers, expenses, and approvals.
- Each audit entry shall include the actor or system process, action, timestamp, relevant business and store, affected record, and outcome.
- Before-and-after values shall be recorded for applicable modifications, excluding secrets and sensitive authentication data.
- Ordinary users shall not be able to edit or delete audit entries.
- Authorized administrators shall be able to search and filter logs by user, action, date, store, and record reference.

> ACCEPTANCE CRITERIA
> A stock adjustment can be traced to its author, reason, approval where applicable, timestamp, and resulting movement record.

FR-28

Data Import and Export

- The system shall allow authorized users to import supported product, supplier, and opening-stock records using documented CSV templates.
- Imports shall validate required fields, data types, identifiers, duplicate records, and store associations before committing changes.
- The system shall provide an error report identifying invalid rows and reasons for rejection.
- Authorized users shall be able to export permitted reports and records to CSV, and PDF where appropriate.
- Exported data shall respect the requesting user's business and store permissions.

> ACCEPTANCE CRITERIA
> An import containing invalid rows produces a clear error report and does not silently create partial or duplicate records.

FR-29

Business Rules, Approvals, and Transaction Integrity

- The system shall enforce configurable approval rules for stock adjustments, write-offs, refunds, discounts, expenses, and other designated high-risk operations.
- Approval thresholds shall be configurable by authorized administrators, with clear rules for amount, quantity, percentage, or transaction type.
- A user shall not approve their own transaction when the applicable policy prohibits self-approval.
- Related financial and inventory updates shall be committed atomically or processed through a documented recoverable workflow that prevents inconsistent final states.
- Failed operations shall return a meaningful error and shall not leave untracked financial or stock changes.

> ACCEPTANCE CRITERIA
> A transaction exceeding an approval threshold cannot complete until the required authorization is obtained; a simulated failure cannot leave a sale saved with an incorrect stock balance.

FR-30

Configuration, System Administration, and Operational History

- Authorized administrators shall be able to manage supported system-wide settings, store settings, transaction numbering rules, tax configurations, notification preferences, and approval policies.
- The system shall provide authorized administrators with operational views of failed jobs, integration failures, and recoverable transaction-processing errors.
- Configuration changes shall be validated, permission-checked, and audited.
- The system shall preserve the configuration or transaction values required to explain historical sales, receipts, taxes, and inventory movements.
- The system shall support documented administrative procedures for recovery, maintenance, and controlled data correction.

> ACCEPTANCE CRITERIA
> An authorized administrator can change a permitted setting, trace the change in the audit log, and verify that historical transactions remain unchanged.

PART B

Non-functional requirements

NFR-01 – NFR-30

Non-functional requirements describe how well, securely, or reliably the system must do it.

> **TARGET-SETTING NOTE**
> The values below are proposed initial service-level targets, not claims about an existing implementation. Confirm them against the expected deployment environment, available budget, network conditions, and anticipated transaction volume before treating them as contractual service levels.

PART B · SECTION 1

1. Performance

NFR-01

Application Response Time

- At the agreed reference load, at least 95% of standard product lookup, stock lookup, and transaction-history requests shall complete within 2 seconds.
- At least 99% of these requests shall complete within 4 seconds.
- Measurements shall exclude user think time and separately identify external service delays.

> **VERIFICATION**
> Automated load tests shall record response-time percentiles against representative datasets.

NFR-02

Search Performance

- Product search by SKU, barcode, or exact identifier shall return results within 300 milliseconds at the 95th percentile under the agreed reference load.
- General product-name and category searches shall return results within 500 milliseconds at the 95th percentile.
- The initial performance test shall include at least 100,000 product or variant records, with a documented path to testing one million records if that scale is required.

> **VERIFICATION**
> Search tests shall measure query latency, result accuracy, and database resource consumption.

NFR-03

Sales Transaction Processing

- At the agreed reference load, at least 95% of ordinary in-store sales shall complete within 2 seconds after payment confirmation, excluding receipt-printer delays and external payment-provider latency.
- A completed sale, its payment records, stock changes, and required transaction records shall be consistent when success is returned.
- Failed transactions shall return an explicit failure result without leaving partial committed changes.

> **VERIFICATION**
> Automated end-to-end tests shall measure completion time and verify sale, payment, and inventory records.

NFR-04

Report Generation Time

- Standard daily, weekly, and monthly reports shall complete within 5 seconds at the 95th percentile under the agreed reference load.
- Reports covering up to one year of data shall complete within 10 seconds at the 95th percentile.
- Large exports exceeding these targets shall use an asynchronous job with a visible status and a retrievable result.

> **VERIFICATION**
> Tests shall use representative historical data and verify both execution time and numerical accuracy.

PART B · SECTION 2

2. Scalability and Capacity

NFR-05

Initial Capacity

- The initial production configuration shall support at least 100 concurrent authenticated users across multiple stores.
- It shall sustain at least 20 completed sales per second for a 15-minute load-test period, provided the configured payment and receipt workflows permit that throughput.
- During the test, at least 95% of ordinary requests shall meet the response-time targets in NFR-01.

> **VERIFICATION**
> A documented load test shall measure throughput, latency, error rates, and database resource usage.

NFR-06

Horizontal Scalability

- The application shall support adding application instances without requiring a redesign of its business logic or database schema.
- Session handling, background jobs, and shared application state shall not depend on a single application instance's local memory.
- The deployment shall support testing at five times the agreed baseline traffic, subject to available infrastructure and database capacity.

> **VERIFICATION**
> A scaling test shall demonstrate that additional application instances can serve requests without introducing duplicate transactions or inconsistent stock balances.

NFR-07

Data Growth and Multi-Store Expansion

- The system shall support adding stores without requiring separate application deployments or duplicated business logic for each store.
- The data model shall distinguish business, store, user, product, inventory, and transaction ownership where applicable.
- The initial architecture shall be tested with at least 100 stores and 100,000 product or variant records.

> **VERIFICATION**
> A test dataset shall demonstrate that adding stores does not expose cross-store data or break existing transaction processing.

PART B · SECTION 3

3. Availability and Reliability

NFR-08

Service Availability

- The production system shall target at least 99.5% monthly availability, excluding only explicitly agreed maintenance windows and documented external dependencies.
- Availability shall be measured using successful synthetic checks against critical workflows, not merely whether the server responds.
- Critical availability failures shall generate an alert within 5 minutes of detection.

> **VERIFICATION**
> Monitoring records shall be used to calculate monthly availability and investigate service interruptions.

NFR-09

Inventory Consistency and Concurrency

- Concurrent sales shall not cause available stock to become negative when negative inventory is prohibited.
- Stock validation and deduction shall be protected by appropriate database transactions, locking, conditional updates, or equivalent concurrency controls.
- Each successful stock movement shall be reflected exactly once in the resulting stock balance.

> **VERIFICATION**
> A concurrent-sale test against one remaining unit shall prove that no more than one sale can consume that unit.

NFR-10

Payment and Transaction Idempotency

- Retrying a payment callback or transaction request with the same idempotency key or external transaction reference shall not create a duplicate successful payment, refund, sale, or stock movement.
- The system shall retain sufficient transaction references to identify and reconcile repeated requests.
- Duplicate requests shall return or expose the original transaction outcome where appropriate.

> **VERIFICATION**
> Replaying the same request at least 10 times shall produce no additional financial charge or duplicate stock deduction.

NFR-11

Failure Isolation and Recovery

- Failure of a non-critical external service, including email or SMS delivery, shall not corrupt committed sales or inventory records.
- Recoverable asynchronous operations shall be queued, retried according to a documented policy, and recorded when they ultimately fail.
- Repeated retries shall not create duplicate business transactions.

> **VERIFICATION**
> Simulated notification-provider failures shall leave completed sales intact and allow delivery to recover without duplicate transactions.

PART B · SECTION 4

4. Backup and Disaster Recovery

NFR-12

Database Backup Frequency

- Automated full backups shall run at least once every 24 hours.
- Where supported by the database platform, point-in-time recovery shall provide a recovery point objective of 15 minutes or less.
- Backups shall be encrypted and protected against unauthorized access and accidental deletion.
- At least one recoverable copy shall be protected independently of the primary database failure domain.

> **VERIFICATION**
> Backup logs shall confirm completion, retention, access protection, and recovery-point capability.

NFR-13

Recovery Time and Recovery Point Objectives

- The initial recovery time objective (RTO) shall be 4 hours or less for a declared major database or application failure.
- The recovery point objective (RPO) shall be 15 minutes or less for supported database recovery scenarios.
- Recovery procedures shall document the responsibilities, dependencies, and steps required to restore the service.

> **VERIFICATION**
> A disaster-recovery exercise shall measure actual restoration time and the amount of data potentially lost.

NFR-14

Backup Restoration Testing

- A restoration test shall be conducted at least once every quarter.
- Each test shall verify database integrity, key business records, stock balances, and the ability to authenticate and execute a test transaction.
- Test outcomes, recovery time, issues, and corrective actions shall be documented.

> **VERIFICATION**
> The quarterly recovery report shall record whether the recovery objectives were achieved and whether corrective action is required.

PART B · SECTION 5

5. Security and Data Protection

NFR-15

Encryption in Transit and at Rest

- All production application traffic carrying credentials or business data shall use HTTPS with TLS 1.2 or higher.
- Sensitive stored data and backups shall use encryption at rest appropriate to the deployment environment.
- Encryption keys and secrets shall not be stored in source code or ordinary application logs.

> **VERIFICATION**
> Deployment and configuration checks shall confirm secure transport, storage encryption, and protected secret management.

NFR-16

Password and Authentication Security

- Passwords shall be hashed using Argon2id or another approved password-hashing algorithm configured according to current security guidance.
- The system shall enforce a documented password and account-recovery policy.
- After five consecutive failed login attempts, the system shall apply a temporary lockout or equivalent progressive throttling mechanism, with recovery controls.

> **ACCEPTANCE CRITERIA**
> Passwords are never recoverable in plaintext, expired or reused reset tokens are rejected, and the configured lockout or throttling policy activates after five consecutive failed attempts.

NFR-17

Staff Authentication and Session Security

- Staff accounts shall require multi-factor authentication (MFA) before accessing production administrative functions.
- Administrative sessions shall expire after 15 minutes of inactivity, unless a documented, risk-based exception is approved.
- The system shall revoke active sessions when an account is deactivated or its credentials are reset, according to the security policy.

> **ACCEPTANCE CRITERIA**
> A staff member cannot access protected administrative functions without completing MFA; an inactive session expires after the configured period; a revoked session cannot perform further protected actions.

NFR-18

Payment Card Security

- The system shall use a payment provider that supports the applicable PCI DSS requirements for its integration model.
- The system shall not store full payment card numbers, card verification values (CVVs), or sensitive authentication data after authorization.
- Payment credentials shall be handled using approved provider-hosted checkout, tokenization, or equivalent controls.

> ACCEPTANCE CRITERIA
> A review of the database, application logs, API responses, and stored payment records finds no prohibited card data; the payment integration's applicable PCI DSS responsibilities are documented.

NFR-19

Application Security and Vulnerability Management

- The system shall address applicable OWASP Top 10 risks and use secure development practices for authentication, authorization, input validation, and transaction processing.
- Automated dependency and static-analysis scans shall run in the development or continuous integration pipeline.
- A security assessment shall be completed before production launch and at least annually thereafter.
- No unresolved critical or high-severity vulnerability shall be accepted for production without documented risk assessment and explicit authorization.

> ACCEPTANCE CRITERIA
> The release pipeline produces security-scan results; the launch assessment records findings and remediation status; release approval confirms that the vulnerability policy has been satisfied.

NFR-20

API Abuse Protection

- Publicly reachable and authenticated APIs shall enforce configurable rate limits appropriate to the endpoint and user role.
- Authentication, password-reset, and other abuse-sensitive endpoints shall have stricter limits than ordinary read operations.
- The system shall return an appropriate rate-limit response when a client exceeds its allowed request rate.
- Rejected requests shall not create duplicate transactions or bypass authorization.

> ACCEPTANCE CRITERIA
> Automated tests exceeding configured limits receive a rate-limit response, while requests within the allowed limit continue to work normally.

PART B · SECTION 6

6. Usability and Accessibility

NFR-21

Responsive Interface

- The application shall support screen widths from 320 pixels to 1920 pixels.
- Core workflows—including login, product search, stock lookup, sales, purchase receiving, and transfer approval—shall remain usable on supported screen sizes.
- The interface shall not require horizontal scrolling for ordinary forms or primary actions at supported viewport sizes, except for explicitly designed data tables.

> ACCEPTANCE CRITERIA
> Automated and manual interface tests at 320 px, 375 px, 768 px, 1024 px, and 1920 px confirm that primary workflows remain accessible and usable.

NFR-22

Accessibility Compliance

- Customer-facing interfaces, where included, and core staff interfaces shall target WCAG 2.2 Level AA.
- Interactive elements shall support keyboard navigation, visible focus indicators, meaningful labels, and accessible error messages.
- Text and interface components shall meet applicable contrast requirements.

> ACCEPTANCE CRITERIA
> An automated accessibility scan and manual keyboard and screen-reader checks shall identify no unresolved critical accessibility blockers in the core workflows.

NFR-23

Checkout and Sales Workflow Efficiency

- A trained cashier shall be able to complete a standard sale involving up to five distinct products without navigating away from the point-of-sale workflow.
- A standard sale shall require no more than six primary interaction stages, excluding payment-provider authentication and exceptional approval steps.
- Product lookup, quantity changes, total calculation, payment recording, and receipt generation shall be available within the same transaction workflow.

> ACCEPTANCE CRITERIA
> In a usability test, at least 90% of participating trained cashiers complete the standard five-product sale without assistance or a workflow-blocking error.

NFR-24

Staff Training and Learnability

- A new cashier shall be able to complete a standard sale after no more than one hour of structured training.
- A new inventory officer shall be able to perform a stock lookup and a permitted stock adjustment after no more than two hours of structured training.
- Training time and task success shall be measured using documented scenarios.

> ACCEPTANCE CRITERIA
> At least 90% of test participants in each role complete their assigned core tasks independently within the specified training period.

PART B · SECTION 7

7. Compatibility and Maintainability

NFR-25

Browser and Device Compatibility

- The web application shall support the latest two stable major versions of Chrome, Edge, Firefox, and Safari at the time of each release.
- Supported mobile browsers shall include current Chrome on Android and Safari on iOS.
- The application shall degrade gracefully when a browser does not support a non-essential feature.

> ACCEPTANCE CRITERIA
> The core workflow test suite passes on every browser in the agreed support matrix, with no release-blocking layout or functional defects.

NFR-26

Automated Testing and API Documentation

- Core business modules—including authentication and authorization, inventory, sales, payments, purchasing, and transfers—shall achieve at least 80% line coverage from automated tests.
- Critical transaction paths shall include integration tests covering success, failure, duplicate requests, and concurrent operations.
- All externally accessible APIs shall have maintained documentation describing authentication, permissions, request and response formats, validation errors, and relevant status codes.

> ACCEPTANCE CRITERIA
> The continuous integration pipeline reports at least 80% line coverage for designated core modules, critical integration tests pass, and API documentation is available for all public endpoints.

NFR-27

Deployment and Rollback

- Production deployments shall use a documented release process that minimizes service interruption.
- The target for routine deployments shall be no more than 60 seconds of user-visible interruption, with zero downtime as the preferred outcome.
- A rollback procedure shall allow the previous compatible application release to be restored within 10 minutes.
- Database changes shall use a backward-compatible migration strategy or a documented recovery procedure.

> ACCEPTANCE CRITERIA
> A deployment rehearsal records interruption duration, and a rollback drill restores the previous compatible release within 10 minutes without corrupting committed transactions.

NFR-28

Centralized Monitoring and Alerting

- Application errors, request latency, database health, background-job failures, and relevant security events shall be captured by centralized monitoring.
- Critical production failures shall trigger an alert within five minutes of detection.
- Monitoring shall avoid recording passwords, authentication tokens, CVVs, or other prohibited sensitive data.
- Alerts shall identify the affected component, severity, occurrence time, and relevant diagnostic reference.

> ACCEPTANCE CRITERIA
> Simulated critical failures trigger an alert within five minutes, and authorized personnel can locate the associated diagnostic records without exposing secrets.

PART B · SECTION 8

8. Compliance, Localization, and Data Integrity

NFR-29

Privacy and Data Protection

- The system shall implement applicable data protection obligations, including the Nigeria Data Protection Act (NDPA) where applicable and GDPR where its territorial scope applies.
- Personal data shall be collected and retained according to documented purposes, lawful grounds, and retention rules.
- Authorized personnel shall be able to process verified requests for personal-data access, export, correction, and deletion or restriction where legally required.
- Eligible requests shall be completed within the applicable statutory deadlines, subject to lawful exceptions and identity verification.
- Deletion workflows shall account for backups, audit requirements, financial-record retention, and other applicable legal obligations.

> ACCEPTANCE CRITERIA
> Documented privacy-request tests demonstrate identity verification, controlled data export, lawful deletion or restriction, and completion within the applicable deadline. No universal 30-day deletion promise shall override a different statutory deadline or a legal retention obligation.

NFR-30

Currency, Localization, and Financial Accuracy

- The system shall support configurable business currency, timezone, date formats, and applicable tax rules.
- If multiple currencies are enabled, each transaction shall retain its original currency and amounts; conversions shall record the exchange rate, rate date, and rounding policy.
- Financial amounts shall use integer minor units where suitable, or an exact decimal representation; binary floating-point arithmetic shall not be used for monetary calculations.
- Financial calculations shall follow documented rounding rules and preserve the original values of completed transactions.
- If a public-facing storefront is enabled, its pages shall support clean URLs, appropriate metadata, and a generated sitemap.

> ACCEPTANCE CRITERIA
> Automated tests confirm correct currency and tax calculations, deterministic rounding, and unchanged historical transaction values after configuration changes. Multi-currency conversions preserve the original amounts and recorded exchange rates.

PART C

Requirement acceptance policy

Before a requirement is marked complete, the team shall:

01 Link the requirement to at least one test case.

02 Record the expected result and actual result.

03 Record whether the test passed or failed.

04 Attach test output or other appropriate evidence.

05 Document any exception, known limitation, or approved deviation.

06 Obtain approval from the designated product owner or reviewer for requirements affecting financial accuracy, inventory integrity, security, or access control.

> **IMPORTANT**
> These acceptance criteria define proposed targets for Version 1.0. Performance, capacity, availability, and recovery targets must be validated against the chosen infrastructure and expected business volume before production commitments are made.

END OF VERSION 1.0